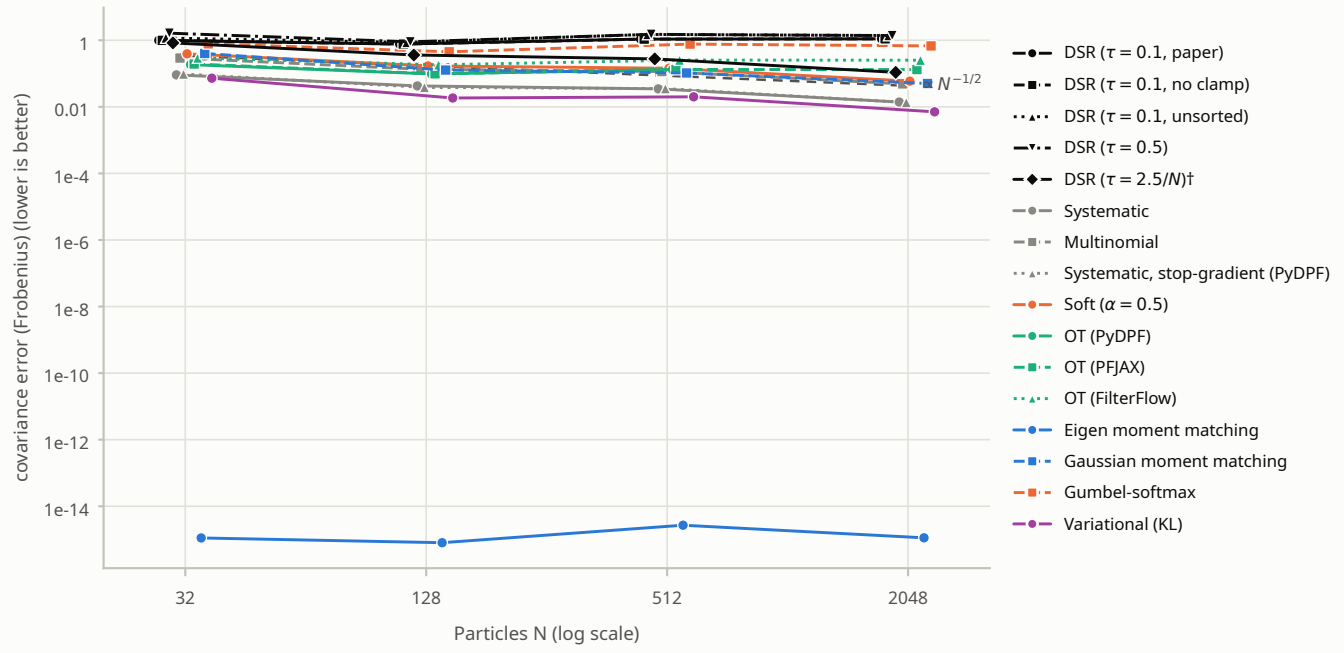
covariance error (Frobenius) (lower is better)
1
0.01
1e-4
1e-6
1e-8
1e-10
1e-12
1e-14
$N^{-1/2}$
32
128
512
2048
Particles N (log scale)
DSR ($\tau = 0.1$, paper)
DSR ($\tau = 0.1$, no clamp)
DSR ($\tau = 0.1$, unsorted)
DSR ($\tau = 0.5$)
DSR ($\tau = 2.5/N$)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft ($\alpha = 0.5$)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)